

Day 00 — Introduction & 30-Day Roadmap

Building a hands-on Security Operations Center home lab from scratch — log collection, detection engineering, attack simulation, and investigation.

Harshkumar Raval • HRTechVerse • SOC Analyst Portfolio

- ELK STACK
- SYSMON
- DETECTION ENGINEERING
- MITRE ATT&CK
- C2 / MYTHIC
- INCIDENT TRIAGE

Overview

The **30 Days SOC Analyst Challenge** by Steven (MyDFIR) is a free, month-long program designed to close the single biggest gap for entry-level SOC candidates: **real, hands-on practical experience**. Each day introduces one new concept or task, building toward a fully functioning home SOC. The core thesis is simple — certifications and theory establish a foundation, but employers hire analysts who can **apply** that knowledge: ingest logs, write detections, investigate alerts, and document findings.

"Certifications and theory are good, but you need to learn how to apply that."
— STEVEN (MyDFIR), 8 yrs in security operations, DFIR consultant

What I will be able to do by the end

- Design and draw a **logical network diagram** for a SOC environment
- Stand up and configure my own **Elasticsearch + Kibana** instance
- Ingest endpoint logs — including **Sysmon** — from Windows hosts
- Simulate, detect, and investigate **malware** and **SSH/RDP brute-force** activity
- Build **detection alerts and dashboards** in Kibana
- Stand up a **Command & Control (C2)** server and detect its activity
- Integrate a **ticketing system** and run end-to-end investigations

The SOC Analyst Workflow This Lab Recreates

Every task in the challenge maps to a stage of the real-world detection-and-response pipeline an analyst lives in daily:



The 4-Week Roadmap

WEEK 1 — FOUNDATION

The ELK Stack & Log Ingestion

Introduction to the Elastic stack. Set up Elasticsearch and Kibana, then ingest logs from endpoints — including **Sysmon** telemetry from Windows hosts. This is the data backbone every later detection depends on.

WEEK 2 — DETECTION

Brute-Force Attacks

Stand up a public-facing **SSH and RDP** server, then build the **alerts and dashboards** that catch brute-force login attempts against it. First taste of detection engineering against live attacker noise.

WEEK 3 — ADVERSARY SIMULATION

Command & Control (C2)

Spin up a personal C2 server using **Mythic** and launch attacks against the public servers. Learn what beaconing and post-exploitation look like from *both* the attacker and defender side.

WEEK 4 — RESPONSE

Ticketing & Investigations

Set up and integrate a **ticketing system**, then run full investigations — taking an alert from detection through triage to a documented, closed case. The capstone that ties the whole pipeline together.

Key Tools & Concepts

- The ELK Stack

 - **Elasticsearch** — search & analytics engine that stores and indexes all log data
- Endpoint Telemetry

 - **Sysmon** — Windows system monitor that logs process creation, network

- **Kibana** — visualization layer where dashboards, alerts, and detection rules are built
 - **Logstash / Agents** — collect, parse, and ship logs from endpoints into Elasticsearch
- connections, and file/registry changes in rich detail

 - This high-fidelity data is what makes attacker behaviour *visible* and maps cleanly to MITRE ATT&CK

Adversary Behaviour Mapped to MITRE ATT&CK

Activity in the Lab	ATT&CK Reference	What the Analyst Looks For
SSH / RDP Brute Force	T1110 Brute Force	Spikes of failed logins from a single source, followed by a success
RDP Remote Access	T1021.001 Remote Services: RDP	Unexpected inbound RDP sessions and lateral movement
Command & Control (Mythic)	TA0011 Command & Control	Beaconing patterns, unusual outbound connections
Malware Execution	T1204 User Execution	Suspicious process trees and child-process spawning in Sysmon

Lab Infrastructure

BUILD DECISION

The challenge is run **in the cloud** because the full stack (ELK + endpoints + attack servers) is resource-intensive. **Vultr** is recommended via a free starter credit, but *any* provider that can spin up a Windows VM works — and an on-prem build (e.g. VirtualBox/VMware) is also valid for anyone with the hardware.

Day 0 Action Checklist

- ✓ Choose the build path: cloud provider (Vultr / Azure / AWS) or on-prem virtualization
- ✓ Create the cloud account and claim available starter credit
- ✓ Sketch the **logical network diagram** — ELK server, Windows endpoint(s), and public attack server
- ✓ Confirm budget guardrails (shut down VMs when idle to preserve credit)
- ✓ Set up this repository to track notes, diagrams, and evidence for all 30 days
- ✓ Commit publicly each day — accountability is the whole point

MY TAKEAWAY

This is where the talk-vs-action line gets drawn. Anyone can list a tool on a resume; far fewer can say "I built the pipeline, generated the attack, wrote the detection, and investigated the alert." Over the next 30 days, that is exactly what this lab produces — and every step lands here as documented, defensible evidence.